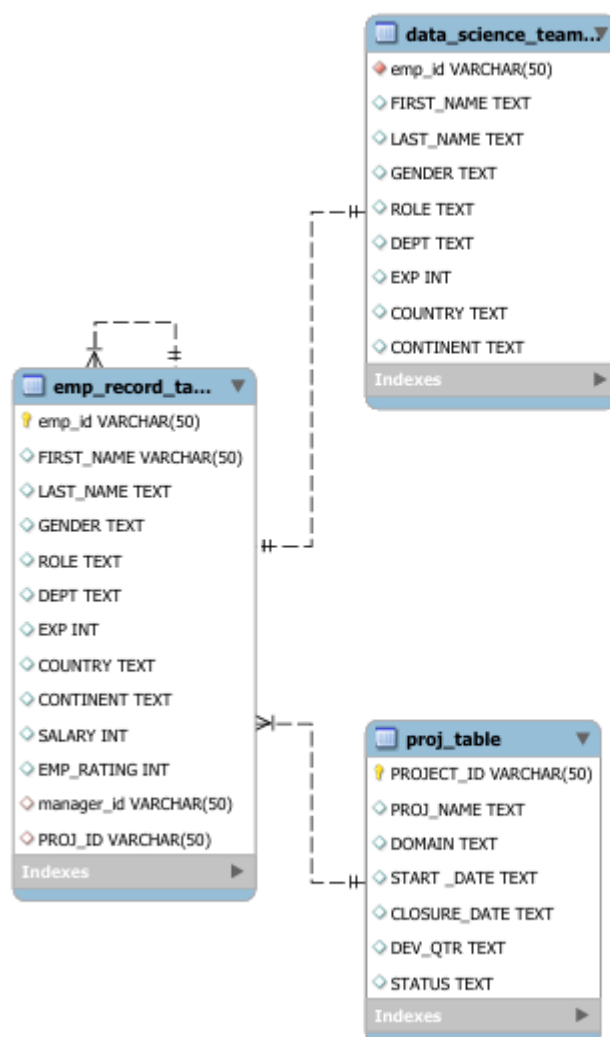


1. Create a database named *employee*, then import **data_science_team.csv**, **proj_table.csv** and **emp_record_table.csv** into the **employee** database from the given resources.

```
1 • create database employee;  
2 • use employee;
```

2. Create an ER diagram for the given **employee** database.



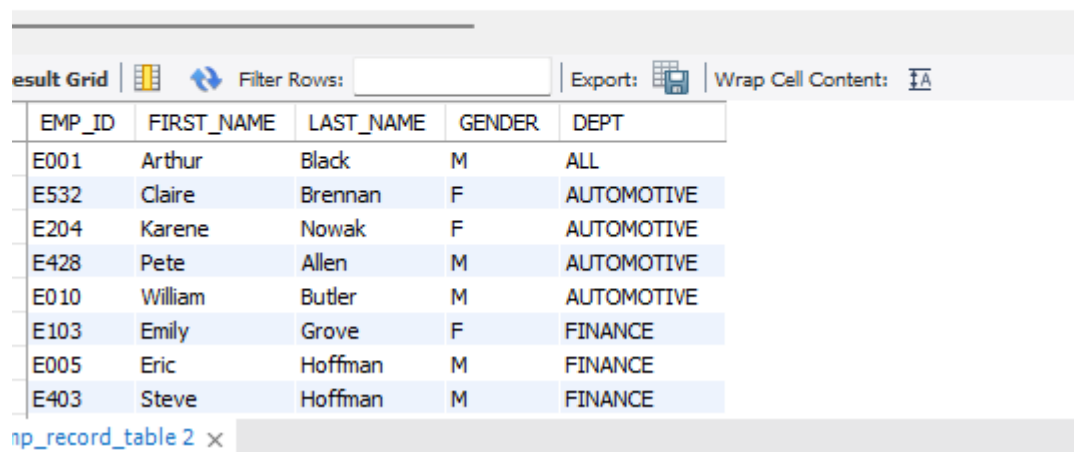
One-to-Many (Self-Referencing): The relationship between **emp_id** and **manager_id** in the **emp_record_table** is a one-to-many relationship. One employee (**emp_id**) can be the manager of multiple employees (**manager_id**), but each employee can have only one manager.

Many-to-One: The relationship between emp_record_table.proj_id and proj_table.project_id is a many-to-one relationship. Multiple employees (emp_record_table.proj_id) can be assigned to the same project (proj_table.project_id), but each project can have multiple employees.

One-to-One: The relationship between emp_record_table.emp_id and data_science_team.emp_id is a one-to-one relationship. Each row in emp_record_table can be linked to at most one matching row in data_science_team. And each row in data_science_team matches exactly one employee in emp_record_table.

3. Write a query to fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, and DEPARTMENT from the employee record table, and make a list of employees and details of their department.

```
3 • SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT
4 FROM emp_record_table
5 ORDER BY DEPT, FIRST_NAME;
```



The screenshot shows a database query result grid. At the top, there are tabs for 'result Grid', a grid icon, a refresh icon, and a 'Filter Rows:' input field. To the right are 'Export:' and 'Wrap Cell Content:' options. Below this is a table with 5 columns: EMP_ID, FIRST_NAME, LAST_NAME, GENDER, and DEPT. The table contains 8 rows of employee data. At the bottom, there is a tab labeled 'emp_record_table 2' with a close button (x).

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPT
E001	Arthur	Black	M	ALL
E532	Claire	Brennan	F	AUTOMOTIVE
E204	Karene	Nowak	F	AUTOMOTIVE
E428	Pete	Allen	M	AUTOMOTIVE
E010	William	Butler	M	AUTOMOTIVE
E103	Emily	Grove	F	FINANCE
E005	Eric	Hoffman	M	FINANCE
E403	Steve	Hoffman	M	FINANCE

4. Write a query to fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPARTMENT, and EMP_RATING if the EMP_RATING is:
 - less than two
 - greater than four
 - between two and four

6

```
7 • SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT, EMP_RATING
8 FROM emp_record_table
9 WHERE EMP_RATING < 2;
```

10

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPT	EMP_RATING
E057	Dorothy	Wilson	F	HEALTHCARE	1
E532	Claire	Brennan	F	AUTOMOTIVE	1
E620	Katrina	Allen	F	RETAIL	1

```
11 • SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT, EMP_RATING
12 FROM emp_record_table
13 WHERE EMP_RATING > 4;
```

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPT	EMP_RATING
E001	Arthur	Black	M	ALL	5
E052	Dianna	Wilson	F	HEALTHCARE	5
E083	Patrick	Voltz	M	HEALTHCARE	5
E204	Karene	Nowak	F	AUTOMOTIVE	5

emp_record_table 4 ×

```
15 • SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT, EMP_RATING
16 FROM emp_record_table
17 WHERE (EMP_RATING BETWEEN 2 AND 4)
```

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPT	EMP_RATING
E005	Eric	Hoffman	M	FINANCE	3
E010	William	Butler	M	AUTOMOTIVE	2
E103	Emily	Grove	F	FINANCE	4
E245	Nian	Zhen	M	RETAIL	2
E260	Roy	Collins	M	RETAIL	3
E403	Steve	Hoffman	M	FINANCE	3
E428	Pete	Allen	M	AUTOMOTIVE	4
E478	David	Smith	M	RETAIL	4

emp_record_table 5 ×

5. Write a query to concatenate the FIRST_NAME and the LAST_NAME of employees in the *Finance* department from the employee table and then give the resultant column alias as NAME.

```
20 ❌ select CONCAT(FIRST_NAME, ' ', LAST_NAME) AS NAME
21     FROM emp_record_table
22     WHERE DEPT= 'FINANCE';
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
NAME				
Eric Hoffman				
Emily Grove				
Steve Hoffman				

6. List each employee with their manager's name

```
SELECT
    e.EMP_ID,
    e.FIRST_NAME      AS employee_name,
    m.FIRST_NAME      AS manager_name
FROM emp_record_table AS e
LEFT JOIN emp_record_table AS m
    ON e.MANAGER_ID = m.EMP_ID;
```

emp_id	employee_name	manager_name
E001	Arthur	NULL
E005	Eric	Emily
E010	William	Pete
E052	Dianna	Patrick
E057	Dorothy	Patrick
E083	Patrick	Arthur
E102	Emily	Arthur

7. Write a query to list down all the employees from the healthcare and finance departments using union. Take data from the employee record table.

```
24 • SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT
25 FROM emp_record_table
26 WHERE DEPT = 'Healthcare'
27
28 UNION
29
30 SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT
31 FROM emp_record_table
32 WHERE DEPT = 'Finance';
```

Result Grid					
		Filter Rows:		Export:	Wrap Cell Content:
	EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPT
▶	E052	Dianna	Wilson	F	HEALTHCARE
	E057	Dorothy	Wilson	F	HEALTHCARE
	E083	Patrick	Voltz	M	HEALTHCARE
	E505	Chad	Wilson	M	HEALTHCARE
	E005	Eric	Hoffman	M	FINANCE
	E103	Emily	Grove	F	FINANCE
	E403	Steve	Hoffman	M	FINANCE

Result 10 x

8. Write a query to list down employee details such as EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPARTMENT, and EMP_RATING grouped by dept. Also include the respective employee rating along with the max emp rating for the department.

```
--
34 • SELECT
35     EMP_ID,
36     FIRST_NAME,
37     LAST_NAME,
38     ROLE,
39     DEPT,
40     EMP_RATING,
41     MAX(EMP_RATING) OVER (PARTITION BY DEPT) AS MAX_RATING
42 FROM
43     emp_record_table;
```

Result Grid							
Filter Rows:							
Export: Wrap Cell Content:							
	EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPT	EMP_RATING	MAX_RATING
▶	E001	Arthur	Black	PRESIDENT	ALL	5	5
	E010	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	2	5
	E204	Karene	Nowak	SENIOR DATA SCIENTIST	AUTOMOTIVE	5	5
	E428	Pete	Allen	MANAGER	AUTOMOTIVE	4	5
	E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	1	5
	E005	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	3	4
	E103	Emily	Grove	MANAGER	FINANCE	4	4
	E403	Steve	Hoffman	ASSOCIATE DATA SCIENTIST	FINANCE	3	4

Result 11 x

9. Write a query to calculate the minimum and the maximum salary of the employees in each role. Take data from the employee record table.

```

SELECT
    ROLE,
    MIN(SALARY) AS MIN_SALARY,
    MAX(SALARY) AS MAX_SALARY
FROM
    emp_record_table
GROUP BY
    ROLE;

```

Result Grid			
Filter Rows:			
Export: Wrap Cell Content:			
	ROLE	MIN_SALARY	MAX_SALARY
▶	PRESIDENT	16500	16500
	LEAD DATA SCIENTIST	8500	9000
	SENIOR DATA SCIENTIST	5500	7700
	MANAGER	8500	11000
	ASSOCIATE DATA SCIENTIST	4000	5000
	JUNIOR DATA SCIENTIST	2800	3000

10. Write a query to assign ranks to each employee based on their experience. Take data from the employee record table.

```

SELECT
    EMP_ID,
    FIRST_NAME,
    LAST_NAME,
    ROLE,
    DEPT,
    EXP,
    DENSE_RANK() OVER (ORDER BY EXP DESC) AS EXPERIENCE_RANK
FROM
    emp_record_table;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPT	EXP	EXPERIENCE_RANK
▶	E001	Arthur	Black	PRESIDENT	ALL	20	1
	E083	Patrick	Voltz	MANAGER	HEALTHCARE	15	2
	E103	Emily	Grove	MANAGER	FINANCE	14	3
	E428	Pete	Allen	MANAGER	AUTOMOTIVE	14	3
	E583	Janet	Hale	MANAGER	RETAIL	14	3
	E612	Tracy	Norris	MANAGER	RETAIL	13	4
	E010	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	12	5
	E005	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	11	6

Result 15

×

11. Write a query to create a view that displays employees in various countries whose salary is more than six thousand. Take data from the employee record table.

- **CREATE VIEW** Employees_Salary_Above_6000 **AS**
SELECT
 EMP_ID,
 FIRST_NAME,
 LAST_NAME,
 ROLE,
 DEPT,
 COUNTRY,
 SALARY
FROM
 emp_record_table
WHERE
 SALARY > 6000;
- **SELECT *** **FROM** Employees_Salary_Above_6000;

	EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPT	COUNTRY	SALARY
▶	E001	Arthur	Black	PRESIDENT	ALL	USA	16500
	E005	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	USA	8500
	E010	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	FRANCE	9000
	E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	USA	7700
	E083	Patrick	Voltz	MANAGER	HEALTHCARE	USA	9500
	E103	Emily	Grove	MANAGER	FINANCE	CANADA	10500
	E204	Karene	Nowak	SENIOR DATA SCIENTIST	AUTOMOTIVE	GERMANY	7500
	E245	Nian	Zhen	SENIOR DATA SCIENTIST	RETAIL	CHINA	6500

Employees_Salary_Above_6000... x

12. Write a nested query to find employees with experience of more than ten years. Take data from the employee record table.

```

SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPT, EXP
FROM emp_record_table
WHERE EXP >= ANY (
    SELECT EXP
    FROM emp_record_table
    WHERE EXP > 10
);

```

	EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPT	EXP
▶	E001	Arthur	Black	PRESIDENT	ALL	20
	E005	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	11
	E010	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	12
	E083	Patrick	Voltz	MANAGER	HEALTHCARE	15
	E103	Emily	Grove	MANAGER	FINANCE	14
	E428	Pete	Allen	MANAGER	AUTOMOTIVE	14
	E003	Janet	King	MANAGER	RETAIL	14

emp_record_table 26 x

13. Write a query to create a stored procedure to retrieve the details of the employees whose experience is more than three years. Take data from the employee record table.

DELIMITER //

- **CREATE PROCEDURE** GetEmployeesWithExperience()

```
BEGIN
    SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPT, EXP, SALARY
    FROM emp_record_table
    WHERE EXP > 3;
END //
```

DELIMITER ;

- **CALL** GetEmployeesWithExperience();

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPT	EXP	SALARY
▶	E001	Arthur	Black	PRESIDENT	ALL	20	16500
	E005	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	11	8500
	E010	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	12	9000
	E052	Dianna	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	6	5500
	E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	9	7700
	E083	Patrick	Voltz	MANAGER	HEALTHCARE	15	9500
	E103	Emily	Grove	MANAGER	FINANCE	14	10500
	E204	Karene	Nowak	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	7500

Result 20

14. Write a query using stored functions in the project table to check whether the job profile assigned to each employee in the data science team matches the organization's set standard.

The standard being:

For an employee with experience less than or equal to 2 years assign 'JUNIOR DATA SCIENTIST',

For an employee with the experience of 2 to 5 years assign 'ASSOCIATE DATA SCIENTIST',

For an employee with the experience of 5 to 10 years assign 'SENIOR DATA SCIENTIST',

For an employee with the experience of 10 to 12 years assign 'LEAD DATA SCIENTIST',

For an employee with the experience of 12 to 16 years assign 'MANAGER'.

```

02 DELIMITER //
03
04 • CREATE FUNCTION employee_job_profile(EXP INT)
05 RETURNS VARCHAR(50)
06 DETERMINISTIC
07 BEGIN
08     DECLARE standard_role VARCHAR(50);
09
10     IF experience <= 2 THEN
11         SET standard_role = 'JUNIOR DATA SCIENTIST';
12     ELSEIF EXP between 2 AND 5 THEN
13         SET standard_role = 'ASSOCIATE DATA SCIENTIST';
14     ELSEIF EXP between 5 AND 10 THEN
15         SET standard_role = 'SENIOR DATA SCIENTIST';
16     ELSEIF EXP between 10 AND 12 THEN

```

```

17         SET standard_role = 'LEAD DATA SCIENTIST';
18     ELSEIF EXP between 12 AND 16 THEN
19         SET standard_role = 'MANAGER';
20     END IF;
21     RETURN standard_role;
22 END //
23 DELIMITER ;
24

```

```

SELECT
    e.emp_id,
    e.first_name,
    e.last_name,
    e.role AS Assigned_Role,
    employee_job_profile(e.EXP) AS Expected_Role,
    CASE
        WHEN e.role = employee_job_profile(e.EXP) THEN 'MATCHES'
        ELSE 'DOES NOT MATCH'
    END AS Role_Match_Status
FROM
    data_science_team e
WHERE
    e.role IN
    ('JUNIOR DATA SCIENTIST', 'ASSOCIATE DATA SCIENTIST', 'SENIOR DATA SCIENTIST', 'LEAD DATA SCIENTIST', 'MANAGER');

```

Result Grid						
		Filter Rows:	Export:		Wrap Cell Content:	
	emp_id	first_name	last_name	Assigned_Role	Expected_Role	Role_Match_Status
▶	E005	Eric	Hoffman	LEAD DATA SCIENTIST	LEAD DATA SCIENTIST	MATCHES
	E010	William	Butler	LEAD DATA SCIENTIST	LEAD DATA SCIENTIST	MATCHES
	E052	Dianna	Wilson	SENIOR DATA SCIENTIST	SENIOR DATA SCIENTIST	MATCHES
	E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	SENIOR DATA SCIENTIST	MATCHES
	E204	Karene	Nowak	SENIOR DATA SCIENTIST	SENIOR DATA SCIENTIST	MATCHES
	E245	Nian	Zhen	SENIOR DATA SCIENTIST	SENIOR DATA SCIENTIST	MATCHES
	E260	Dev	Collins	SENIOR DATA SCIENTIST	SENIOR DATA SCIENTIST	MATCHES
Result 27 x						

15. Create an index to improve the cost and performance of the query to find the employee whose FIRST_NAME is 'Eric' in the employee table after checking the execution plan.

```
create index idx_first_name on emp_record_table(FIRST_NAME);
explain select EMP_ID, FIRST_NAME, LAST_NAME
from emp_record_table
where FIRST_NAME = 'Eric';
```

	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Ext
▶	1	SIMPLE	emp_record_table	NULL	ALL	NULL	NULL	NULL	NULL	19	10.00	Usin

16. Write a query to calculate the bonus for all the employees, based on their ratings and salaries (Use the formula: 5% of salary * employee rating).

```
6 • select EMP_ID, FIRST_NAME, LAST_NAME,
7       ROLE, DEPT, EXP, SALARY, EMP_RATING,
8       (0.05*SALARY) * EMP_RATING AS Bonus
9 from emp_record_table;
```

Result Grid								
		Filter Rows:	Export:		Wrap Cell Content:			
ID	FIRST_NAME	LAST_NAME	ROLE	DEPT	EXP	SALARY	EMP_RATING	Bonus
▶	Arthur	Black	PRESIDENT	ALL	20	16500	5	4125.00
	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	11	8500	3	1275.00
	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	12	9000	2	900.00
	Dianna	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	6	5500	5	1375.00
	Dorothy	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	9	7700	1	385.00
	Patrick	Voltz	MANAGER	HEALTHCARE	15	9500	5	2375.00
	Emily	Grove	MANAGER	FINANCE	14	10500	4	2100.00

17. Write a query to calculate the average salary distribution based on the continent and country. Take data from the employee record table.

- **SELECT**
continent,
country,
AVG(salary) AS avg_salary
FROM
emp_record_table
GROUP BY
continent,
country
ORDER BY
continent,
country;

Result Grid   Filter Rows:			
	continent	country	avg_salary
▶	ASIA	CHINA	6500.0000
	ASIA	INDIA	6166.6667
	EUROPE	FRANCE	9000.0000
	EUROPE	GERMANY	7600.0000
	NORTH AMERICA	CANADA	7000.0000
	NORTH AMERICA	USA	9440.0000
	SOUTH AMERICA	COLOMBIA	5600.0000